

Jayasurya T Appu

Backend Engineer | Node.js | Microservices

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SUMMARY

Backend Engineer specializing in scalable microservices and event-driven architectures using Node.js and Python. Proven track record of optimizing system performance by 40–50% through efficient caching and database design. Experienced in building production-grade backend systems handling high-throughput workloads.

SKILLS

Backend & Systems: Node.js, Express.js, Python (Django, Flask), REST APIs, Microservices, gRPC, Web Sockets

Databases & Caching: MongoDB, SQL, Redis (Caching)

Message Queues: Kafka

DevOps & Infrastructure: Docker, Docker Compose, Nginx, GCP

Frontend (Supporting): React.js, Angular

Tools: Git, Postman, API Testing, Debugging

EXPERIENCE

Software Engineer

Carclenx (Pexa)

July 2025 – Present

- Designed and scaled backend systems powering internal tools and customer-facing platforms, handling **~1000 requests/sec** in production.
- Built Worker & Staff Analytics system**, reducing service cancellation rates by **~80%** through data-driven task assignment.
- Engineered automated Payroll Processing system** for **~200 employees (100 weekly + 100 monthly)** using message queues, reducing manual accounting workload by **~70–80%**.
- Migrated WhatsApp automation from third-party dependency to an in-house system**, reducing external costs and driving a **~20% increase in lead generation**.
- Stabilized legacy backend systems** by fixing exception handling and server issues, reducing downtime and preventing missed requests.
- Optimized database queries and backend workflows, improving API response times by **30–50%**.
- Led a team of 2 developers, managing **task distribution, code reviews, and delivery**.

Junior Software Engineer

Quest Innovative Solution

Dec 2022 – Sept 2024

- Designed and developed **20+ REST APIs**, improving backend efficiency and reducing manual processing workflows.
- Built backend services using **Node.js** and **Python (Django)** for business-critical applications.
- Optimized backend performance through efficient query design and logic improvements.
- Integrated frontend applications (**React, Angular**) with backend APIs, ensuring **stable and efficient data flow**.

- Participated in **testing, debugging, and production deployments** across multiple projects.

Full Stack Developer

Carclenx (Pexa)

Jan 2022 – June 2022

- Developed secure RESTful APIs and integrated third-party services.
- Built admin dashboards with Angular, ensuring **real-time backend** integration.
- Improved application reliability by ensuring consistent frontend–backend communication.

Software Developer

Paco Studios (Freelance / Startup Work)

2024 – Present

- Delivered **production-grade applications used by real clients**, handling end-to-end backend architecture and deployment.
- Led **end-to-end development**, including requirement gathering, backend architecture, API development, and deployment.
- Designed and deployed **scalable backend systems** for client projects using Node.js and MongoDB.
- Built and deployed client websites:
 - lorrygopvt.com, poornaya.com , pacostudios.org

PROJECTS

Microservices E-commerce Backend System

- Designed and implemented a **microservices-based backend architecture** using Node.js and Express, with service-level separation and **gRPC for efficient inter-service communication**.
- Built an **event-driven system using Kafka** and integrated **Redis caching**, reducing redundant database queries and improving overall API performance.
- Containerized services using **Docker & Docker Compose** and configured **Nginx as a reverse proxy**, enabling scalable, consistent deployment and centralized request routing.
- Deployed and managed a **scalable production setup using MongoDB**, ensuring efficient data persistence and system reliability

AI Chat System with RAG (Retrieval-Augmented Generation)

- Developed a backend system using **FastAPI, ChromaDB, and SentenceTransformers** for semantic search
- Implemented **vector embedding storage and retrieval**, enabling context-aware responses
- Integrated a **local LLM (Mistral)** for generating intelligent answers based on retrieved data
- Designed the system to handle **document-based querying (PDF → embeddings → response)**
- Built an architecture for **scalable AI-assisted chatbot systems**

EDUCATION

Bachelor of Technology (B.Tech)

Computer Science

2017 – 2021

Coursework completed